



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

CAPSTONE PROJECT PRESENTATION

Cloud Based IaaS Virtual Infrastructure Provisioning and Auto Scaling System

CSA1522-Cloud Computing and Big Data Analytics

Guided By
Dr. P.Rajaram

Presented By
M. Meghana (192525435)
R. Pranathi (192525076)
M.Divyasree(192524098)



TABLE OF CONTENTS

- ✓ Introduction
- ✓ Problem statement
- ✓ Objectives
- ✓ Literature Review
- ✓ Methodology
- ✓ Module 1: VM Provisioning & Resource Allocation
- ✓ Module 2: Auto-Scaling & Load Balancing Engine
- ✓ Module 3: Monitoring & Cloud Resource Management
- ✓ Implementation
- ✓ Results & Discussion
- ✓ Challenges & Limitations
- ✓ Future Scope
- ✓ Conclusion
- ✓ References



INTRODUCTION

- ❖ **IaaS** provides on-demand virtual computing resources such as VMs, storage, and networking through the cloud.
- ❖ Virtualization allows multiple VMs to efficiently share the available physical infrastructure.
- ❖ Changing workloads require infrastructure to scale dynamically to maintain application performance.
- ❖ The proposed system automates VM provisioning and auto-scaling to improve resource utilization, scalability, and cost efficiency.



PROBLEM STATEMENT

- ❖ Manually managing virtual infrastructure becomes challenging when application workloads change frequently and unpredictably. Administrators need to continuously observe resource utilization and decide when to create, configure, or remove virtual machines.
- ❖ During high workload conditions, insufficient computing resources may lead to increased response time, reduced application performance, and service interruptions. On the other hand, allocating more resources than required during low workload periods results in underutilized virtual machines and unnecessary infrastructure costs.



OBJECTIVES

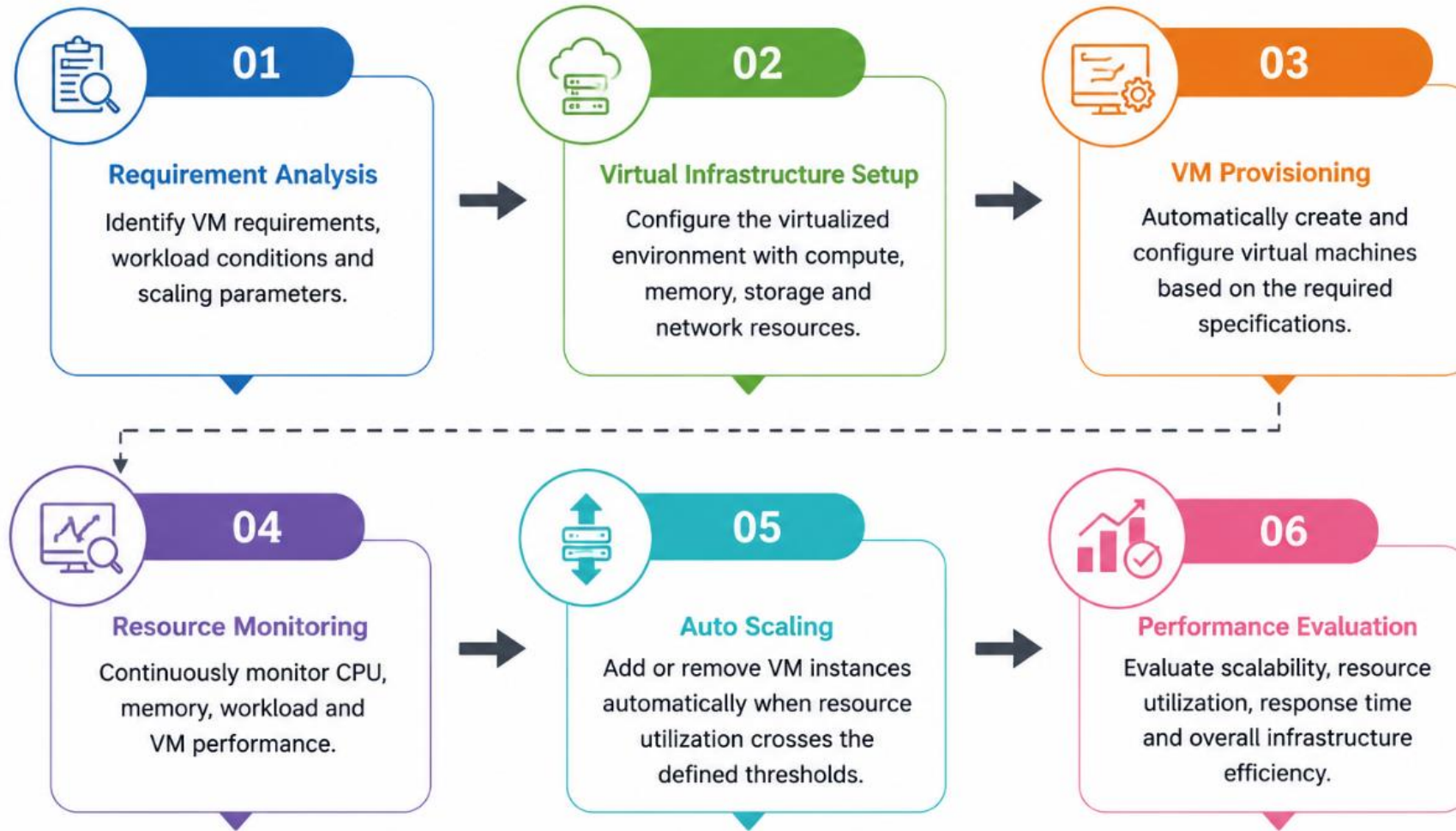
- ❖ Automate the creation and configuration of virtual infrastructure.
- ❖ Establish a mechanism for workload-aware resource adjustment.
- ❖ Maintain application availability during changing workloads.
- ❖ Minimize unnecessary allocation of computing resources.
- ❖ Provide centralized monitoring of virtual infrastructure.
- ❖ Support efficient and scalable IaaS resource management.



LITERATURE REVIEW

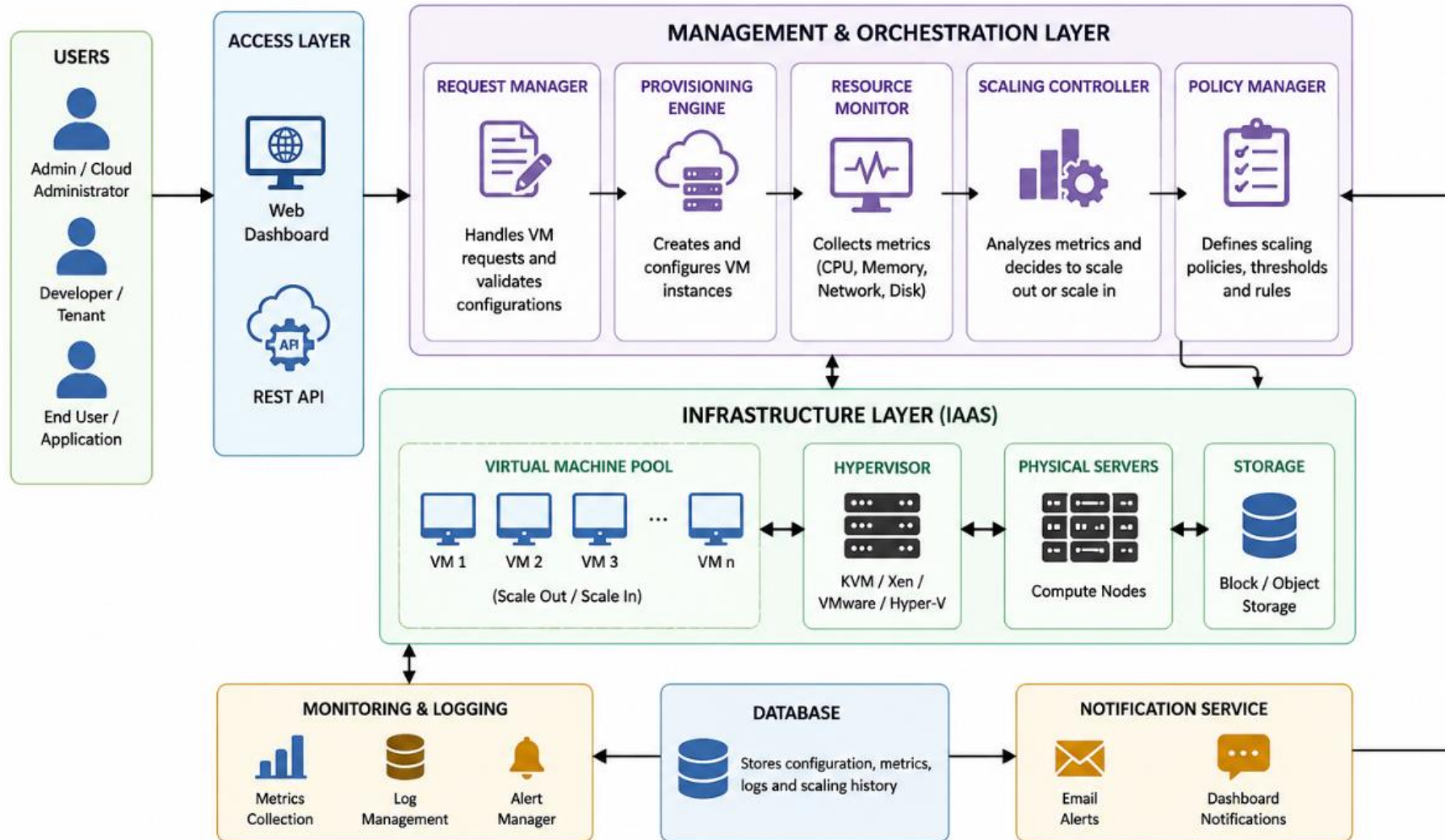
Year	Research Focus	Approach	Limitation	Gap Addressed by Our Project
2023	Online service provisioning in cloud environments	Uses dynamic VM provisioning and scheduling to handle uncertain user requests.	Future workload demand is difficult to predict accurately. (ScienceDirect)	Provides automated VM provisioning with workload-based scaling.
2024	Auto-scaling techniques in cloud computing	Reviews threshold-based, queuing, time-series, and ML-based scaling techniques.	Existing approaches face challenges in balancing cost, performance, and resource utilization. (MDPI)	Combines resource monitoring with practical threshold-based auto-scaling.
2024	Q-learning based auto-scaling for cloud services	Uses Q-learning to dynamically provision VM resources for big-data workloads.	Requires learning-based decision mechanisms and can be computationally complex. (ScienceDirect)	Provides a simpler automated scaling mechanism suitable for IaaS VM environments.
2025	Horizontal VM autoscaling in hybrid cloud	Studies scaling VM instances across public, private, and hybrid cloud environments.	IaaS-level VMs face challenges in achieving rapid elasticity and maintaining QoS. (MDPI)	Implements automatic VM addition/removal based on resource utilization.
2026	ML-based autoscaling for elastic cloud applications	Classifies ML-based reactive, predictive, horizontal, vertical, and hybrid scaling approaches.	Telemetry delays, scaling coordination, and joint cost-SLO-energy optimization remain challenging. (MDPI)	Focuses on an implementable IaaS provisioning and auto-scaling workflow with measurable performance.

METHODOLOGY





SYSTEM ARCHITECTURE





MODULE 1: VM Provisioning & Resource Allocation

- ❖ Receives user requests for virtual machines through the cloud portal.
- ❖ Authenticates users and verifies resource requirements.
- ❖ Automatically provisions Virtual Machines (VMs) using virtualization technology.
- ❖ Allocates CPU, RAM, storage, and network resources based on workload.
- ❖ Continuously manages and optimizes resource allocation for efficient utilization.

```
divya_sree_m@LAPTOP-EDH45F4T:~/devstack/devstack$ openstack server list --all-projects
```

ID	Name	Status	Networks	Image	Flavor	Project ID
c448dca4-7c8a-4878-8ee8-dfb11ded7838	vm1	ACTIVE	private=10.0.0.24, fdc5:8689:3cae:0:f816:3eff:fe7f:c240	N/A (booted from volume)	m1.tiny	3a38c3c2ff4b4d53a80a37653a047f6a

Figure 1.1: OpenStack Nova VM Provisioning Architecture.

```
divya_sree_m@LAPTOP-EDH45F4T:~/devstack/devstack$ openstack hypervisor list
```

ID	Hypervisor Hostname	Hypervisor Type	Host IP	State
a78710d8-fb4d-43e7-b8a2-d6d1a8454a22	LAPTOP-EDH45F4T.localdomain	QEMU	172.21.30.246	up

Figure 1.2: VM1 Successfully Provisioned and Running.



MODULE 2: Auto-Scaling & Load Balancing Engine

- ❖ Monitors CPU, memory, and network utilization.
- ❖ Uses predefined thresholds for scaling decisions.
- ❖ Demonstrates OpenStack's capability to integrate with Heat and Octavia for auto-scaling and load balancing.
- ❖ Distributes traffic using OpenStack networking architecture.
- ❖ Improves scalability and availability.

```
divya_sree_m@LAPTOP-EDH45F4T:~/devstack/devstack$ openstack network list
```

ID	Name	Subnets
853ec43d-13ae-44a0-983e-92958b024417	shared	fa3dd3bf-d218-4fb7-95ac-fb3235d74c3f
8ee45ca0-4bbb-4b17-b3fa-cf4cfeaa0cf1	public	2e2de05f-d2cb-4fe7-b11e-d901b4f9ed85, 660731f8-7c2d-4211-ba27-84e22e247f66
d603f001-84c7-414e-979d-ecb1279479d1	private	4ef8330d-0686-4fee-9626-21cd4fedfaef, b7803708-d316-45f7-a396-5a750fb9c821

Figure 2.1: OpenStack Load Balancer Architecture.

```
divya_sree_m@LAPTOP-EDH45F4T:~/devstack/devstack$ openstack router list
```

ID	Name	Status	State	Project	HA
06fff277-92f6-45e7-9168-b1a6e4639cf9	router1	ACTIVE	UP	3a38c3c2ff4b4d53a80a37653a047f6a	True

Figure 2.2: Network Configuration Supporting Load Balancing.



MODULE 3: Monitoring & Cloud Resource Management

- ❖ **Resource Monitoring** – Continuously monitors CPU, RAM, storage, and network usage of virtual machines.
- ❖ **Performance Tracking** – Checks the performance and health of active VMs in real time.
- ❖ **Threshold Detection** – Identifies when resource usage becomes too high or too low.
- ❖ **Resource Management** – Provides usage information to the auto-scaling engine for scaling decisions.
- ❖ **Cost Optimization** – Helps avoid unnecessary resource usage and improves overall cloud efficiency.

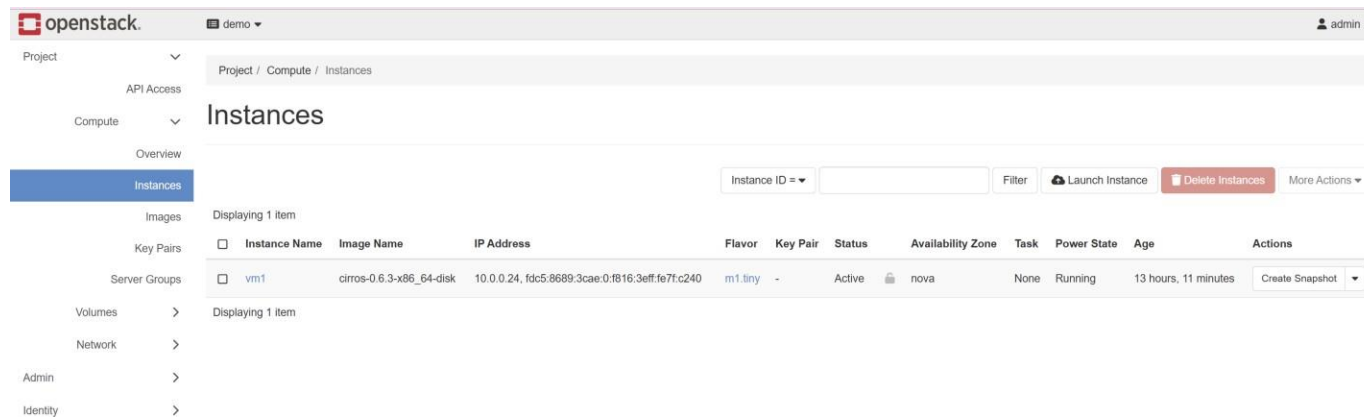


Figure 3.1: Horizon Monitoring Dashboard.

```
divya_sree_m@LAPTOP-EDH45F4T:~/devstack/devstack$ openstack compute service list
```

ID	Binary	Host	Zone	Status	State	Updated At
6130d750-78d2-41b1-b064-22a2ebbc68eb	nova-scheduler	LAPTOP-EDH45F4T	internal	enabled	up	2026-08-20T05:28:47.000000
32a23a33-9871-4a00-ae94-2bc8f9caad0b	nova-conductor	LAPTOP-EDH45F4T	internal	enabled	up	2026-08-20T05:28:47.000000
74377916-8a31-4d0b-9da2-6529c5ae6cbf	nova-conductor	LAPTOP-EDH45F4T	internal	enabled	up	2026-08-20T05:29:11.000000
cb3e7d3b-0b1b-4ee4-98a6-30d51a6321de	nova-compute	LAPTOP-EDH45F4T	nova	enabled	up	2026-08-20T05:28:48.000000

Figure 3.2: OpenStack Compute Services Status.

divya_sree_m@LAPTOP-EDH45F4T:~/devstack/devstack\$ openstack server list --all-projects

ID	Name	Status	Networks	Image	Flavor	Project ID
c448dca4-7c8a-4878-8ee8-dfb11ded7838	vm1	ACTIVE	private=10.0.0.24, fdc5:8689:3cae:0:f816:3eff:fe7f:c240	N/A (booted from volume)	m1.tiny	3a38c3c2ff4b4d53a80a37653a047f6a

Figure 3.3: OpenStack Service Monitoring.



IMPLEMENTATION

- ❖ User authentication using Keystone
- ❖ VM provisioning using Nova
- ❖ Resource allocation (CPU, RAM, Storage)
- ❖ Network configuration using Neutron
- ❖ Storage attachment through Cinder
- ❖ Resource monitoring using Horizon Dashboard
- ❖ Successful deployment and activation of VM1



RESULTS & DISCUSSION

- ❖ Virtual infrastructure is provisioned quickly and efficiently.
- ❖ The deployed infrastructure successfully supports scalable VM provisioning and resource management.
- ❖ Resource utilization is optimized, reducing idle resources.
- ❖ Infrastructure costs are minimized through automatic scaling.
- ❖ The system provides high availability and reliable cloud services.



CHALLENGES & LIMITATIONS

- ❖ Cloud security and data privacy need continuous protection.
- ❖ Network latency may affect application performance.
- ❖ DevStack experienced an Out-of-Memory (OOM) issue in WSL, which was resolved by increasing swap memory and restarting MySQL and OpenStack services.
- ❖ Initial cloud setup and configuration are complex.
- ❖ Sudden workload spikes may cause temporary scaling delays.



FUTURE SCOPE

- ❖ Integrate AI for predictive auto-scaling decisions.
- ❖ Support multi-cloud deployment for better reliability.
- ❖ Use Docker and Kubernetes for container orchestration.
- ❖ Implement serverless computing for improved efficiency.
- ❖ Enhance security using Zero Trust and AI-based monitoring.



CONCLUSION

- ❖ Successfully deployed OpenStack DevStack on Ubuntu WSL.
- ❖ Auto-scaling improves resource utilization and performance.
- ❖ Load balancing ensures continuous service availability.
- ❖ Cloud resources are managed efficiently and cost-effectively.
- ❖ The solution provides a scalable, reliable, and flexible IaaS environment.



REFERENCES

- Rout, S.K., Ravinda, J.V.R., Meda, A., Mohanty, S.N. and Kavididevi, V., 2023. A dynamic scalable auto-scaling model as a load balancer in the cloud computing environment. EAI Endorsed Transactions on Scalable Information Systems, 10(5), pp.1-7.
- Chouliaras, S. and Sotiriadis, S., 2023. An adaptive auto-scaling framework for cloud resource provisioning. Future Generation Computer Systems, 148, pp.173-183.
- Quattrocchi, G., Incerto, E., Pincioli, R., Trubiani, C. and Baresi, L., 2024. Autoscaling solutions for cloud applications under dynamic workloads. IEEE Transactions on Services Computing, 17(3), pp.804-820.
- Betti Pillippuge, T.L., Khan, Z. and Munir, K., 2025. Horizontal autoscaling of virtual machines in hybrid cloud infrastructures: current status, challenges, and opportunities. Encyclopedia, 5(1), p.37.
- Catillo, M., Villano, U. and Rak, M., 2023. A survey on auto-scaling: how to exploit cloud elasticity. International Journal of Grid and Utility Computing, 14(1), pp.37-50.

Thank
you!